

Claude Code Skills Reference

Day 2 Handout - Extensions (July 2026 edition)

Skills are a cross-platform standard: the same SKILL.md runs in Claude Code, Claude.ai, the Agent SDK, and the Developer Platform.

Skill Anatomy

```
your-skill-name/ <- kebab-case folder
SKILL.md <- frontmatter + instructions (case-sensitive name!)
references/ <- large docs, templates, checklists
checklist.md
scripts/ <- computation scripts
parse-data.py
```

Locations: project `.claude/skills/` (shared via git) | global `~/.claude/skills/` | bundled inside plugins.

SKILL.md Frontmatter - Current Format (July 2026)

```
---
name: csv-cleaner
description: >
Transforms messy CSV files into clean spreadsheets.
Use when the user says "clean CSV", "fix headers", "format data",
"organise spreadsheet", or "tidy this data". Also activate when the
user uploads a .csv and asks for cleaning.
Do NOT use for PDFs, Word documents, or image files.
disable-model-invocation: false
allowed-tools: Read, Grep, Glob, Write
---
```

Field	Purpose
name	Kebab-case identifier; also the <code>/name</code> slash trigger
description	THE TRIGGER - Claude reads this to decide whether to load the skill. Written for the model, not for humans
disable-model-in vocation	<code>true</code> = manual-only (<code>/name</code>); use for side-effect skills: <code>/deploy</code> , <code>/commit</code> , <code>/publish</code>
allowed-tools	Restrict the skill's tool access (least privilege - review skills get Read, Grep, Glob only)

Description Rules (the #1 failure point)

- Be **pushy** - Claude is conservative about activation
- 5-7+ explicit trigger phrases
- Negative boundaries: "Do NOT use for..."
- Third person ("Transforms..." not "I can...")
- Under 300 words

Body Template

Target **1,500-2,000 words**; push bulk into [references/](#) (progressive disclosure - you pay tokens only when the skill fires).

```
## Overview
One paragraph: what this does and when it activates.

## Workflow
1. [One imperative action per step]
2. ...

## Output Format
File type, structure, sections, tone.

## Edge Cases
- Missing input -> ask for clarification
- Ambiguous request -> state assumptions, proceed

## Examples
### Happy path (input -> expected output)
### Edge case (unusual input -> handling)

## Gotchas
Document Claude's observed failure points here over time.
```

Add an anti-hallucination rule to review-type skills: *"If no issues are found, explicitly state 'No issues found'."*

The One-Level-Deep Rule

References must not link to other references - Claude truncates nested reads.

Do	Don't
SKILL.md -> references/guide.md	SKILL.md -> references/index.md -> references/guide.md
Compress a 50-page guide to 2 pages	Paste the whole guide into references/

5 Skill Failure Modes

#	Mode	Symptom	Fix
1	Silent	Never fires	More trigger phrases; be pushy
2	Hijacker	Fires on wrong requests	"Do NOT use for..." boundaries
3	Drifter	Wrong output	Specific, testable instructions
4	Fragile	Breaks on edge cases	Explicit "If X -> Y" rules
5	Overachiever	Unsolicited extras	"Output ONLY X, nothing else"

The Eval Loop (Lab 2C)

- 1. **Test set** - 5 should-trigger + 5 should-NOT-trigger prompts in [tests/eval-prompts.md](#)

2. **Run & score** - did the skill's format appear? Trigger rate = your metric
3. **A/B check** - rename SKILL.md off, `/clear`, same prompt; compare structure/consistency
4. **Optimize** - feed real failures back into the description; retest
5. **Rerun after every model update** - trigger behavior drifts

Keep / optimize / retire: capability-uplift skills (teach something new) hold their edge; workflow/preference skills (enforce YOUR format) get overtaken by model progress - retire when raw Claude matches them.

Hygiene & Tooling

- `/skills` - inventory + per-skill token cost of descriptions (always in context)
- `/doctor` (July 2026) - flags **unused skills** automatically
- `claude plugin init` - scaffold skills inside a distributable plugin
- Descriptor cost: every enabled skill's description rides in every prompt. Tighten ruthlessly (~9 tokens beats ~45).
High-frequency skills: auto-invoke on. Rare skills: `disable-model-invocation: true` or remove.

Design Checklist

- ☐ All four frontmatter fields present (name, description, disable-model-invocation, allowed-tools)
- ☐ 5+ trigger phrases + negative boundaries
- ☐ `allowed-tools` = minimum needed
- ☐ Output format explicitly defined
- ☐ 2+ concrete examples (happy path + edge case)
- ☐ References one level deep only
- ☐ Body 1,500-2,000 words max
- ☐ Eval test set exists and was run
- ☐ Gotchas section started

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